

Instructions:

1. Attempt all questions.
2. Write your answers on your own paper.
3. All necessary working must be shown.
4. Marks will be deducted for careless or badly arranged work.

QUESTION ONE : (23 MARKS)

a) Write $1.\dot{3}\dot{2}7\dot{4}$ as a rational number

1

b) Evaluate $\sqrt[3]{\frac{8.3 \times 4.1}{0.2 + 5.4 \div 1.3}}$ correct to 3 significant figures

2

c) If $\sqrt{800} = x\sqrt{2}$ find the value of x

2

d) Simplify

i) $ab + 2b - 3ab + 8b$

ii) $8 - 4(2y + 1) + y$

5

e) Simplify $\frac{(x^2)^{-3} \times (y^3)^2}{x^{-1} \times y^4}$

2

f) Expand and simplify

i) $(2y - 3)(y + 5)$

ii) $(2x + 3)^2$

iii) $(5a - b)(5a + b)$

9

iv) $(\sqrt{6} + 3\sqrt{2})(\sqrt{6} - 3\sqrt{2})$

v) $(\sqrt{5} + \sqrt{2})^2$

g) Express $\frac{2\sqrt{3}}{\sqrt{5}}$ with a rational denominator

2

QUESTION TWO : (51 MARKS)

a) Factorise fully

i) $4x^2y - 2xy$

ii) $xm + 5x + 7m + 35$

iii) $m^2 - 9m + 18$

iv) $4x^2 + 4x - 3$

v) $3x^2 - 12y^2$

vi) $8 + 27y^3$

13

b) Simplify

i) $\frac{x^2}{5x^2 - 2x}$

ii) $\frac{t^2 + t - 2}{t^2 + 5t + 6}$

iii) $\frac{2xy + 2x - 6 - 6y}{4x^2 - 16x + 12}$

iv) $\frac{3}{b+2} \times \frac{b^2 + 2b}{6a - 3}$

v) $\frac{3ab^2}{5xy} \div \frac{12ab - 6a}{x^2y + 2xy^2}$

vi) $\frac{p-3}{6} + \frac{p+2}{2}$

vii) $\frac{2}{x^2 - 4} - \frac{3}{x + 2}$

22

c) Simplify

i) $\sqrt{12} - \sqrt{27}$

ii) $2\sqrt{8} - \sqrt{18}$

iii) $4\sqrt{48} + 3\sqrt{147} + 5\sqrt{12}$

11

d) Find a and b if $\frac{\sqrt{3} - 4}{2 + 3\sqrt{3}} = a + b\sqrt{3}$

5

QUESTION THREE : (38 MARKS)

a) Solve the following

i) $2(3x + 7) = 6 - (x - 1)$

ii) $3y - 4 > 5y + 12$

iii) $-1 \leq 2x + 3 \leq 5$

iv) $\frac{a-2}{3} < 2 + \frac{3a}{4}$

v) $|2x + 1| = 3x - 2$

vi) $|2t - 3| \geq 5$

19

b) Solve giving answers in simplified surd form where necessary

i) $3a^2 - 14a + 8 = 0$

ii) $4x^2 + 12x + 1 = 0$

8

c) If $x = 2$ is one solution to $x^2 - ax - 2 = 0$, find the other solution.

3

d) Solve the simultaneous equations

i) $3a + 2b = 5$
 $2a - b = -6$

ii) $y = x^2 + 4x$
 $2x - y - 1 = 0$

8

QUESTION FOUR : (15 MARKS)

a) Find correct to 3 decimal places

i) $\sin 38^{\circ}25'$

ii) $\tan 125^{\circ}36'$

2

b) Find θ in degrees and minutes if θ is acute

i) $\cos \theta = 0.827$

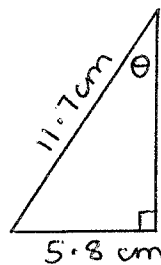
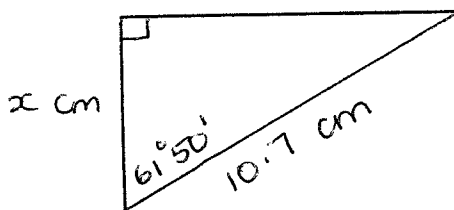
ii) $\sin \theta = \frac{10}{14}$

2

c)

i) Find x to one decimal place

ii) Find θ in degrees and minutes



4

d) The angle of depression from the top of an 80 metre building to a car below is $61^{\circ}29'$. How far is the car from the building to one decimal place ?
(Draw a diagram.)

3

e) A girl rides a motorbike through her property, starting at her house. If she rides south for 1.3 km, then rides west for 2.4 km, what is her bearing from the house, to the nearest degree ?
(Draw a diagram.)

4

QUESTION FIVE : (33 MARKS)

a) Find the exact values of

i) $\sin 60^\circ + \cos 60^\circ$ ii) $\cos^2 45^\circ$ iii) $2 \operatorname{cosec} 30^\circ$

iv) $\frac{\tan 45^\circ + \tan 60^\circ}{1 - \tan 45^\circ \tan 60^\circ}$ v) $\sin 225^\circ$ vi) $\cos 390^\circ$

vii) $\tan(-135^\circ)$ viii) $\sin 240^\circ$

18

b) If $\cos x = \frac{-3}{8}$ and $\sin x < 0$, find the exact value of $\tan x$

3

c) Find the value of θ to the nearest minute for $0^\circ \leq \theta \leq 360^\circ$ for which

i) $\sin \theta = 0.463$ ii) $\cos \theta = -0.237$ iii) $2 \sin \theta = \sqrt{3}$

iv) $\tan 2\theta = \sqrt{3}$ v) $\cos^2 \theta = \frac{1}{2}$

12

QUESTION SIX : (38 MARKS)

a) Simplify

i) $\sin \theta \cot \theta$

ii) $3 + 3 \tan^2 \theta$

iii) $\cot \theta - \cot \theta \cos^2 \theta$

8

b) Prove that

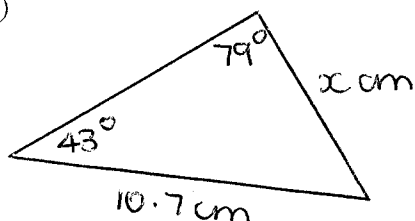
i) $\frac{1 - \sin^2 \theta \cos^2 \theta}{\cos^2 \theta} = \tan^2 \theta + \cos^2 \theta$

ii) $\frac{\cos \theta (\sin \theta + \cos \theta)}{(1 + \sin \theta)(1 - \sin \theta)} = 1 + \tan \theta$

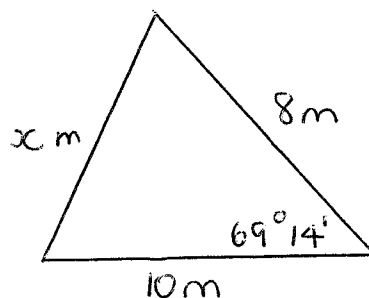
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c) Find x to one decimal place

i)

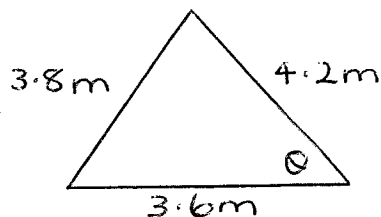


ii)



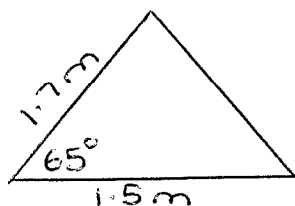
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d) Find θ to the nearest minute.



3

e) Find the area of the sail below, correct to one decimal place.



2

f) Sam drove from his home for 200km on a bearing of 040° , then drives on a bearing of 157° for 345km.

i) Draw a diagram showing this information.

2

ii) How far from home is Sam to the nearest km?

3

iii) What is the bearing of Sam from his home to the nearest degree?

3

g) Sketch the graph of $y = \cos x$ for $0^\circ \leq x \leq 360^\circ$

3

YR 11 2U SOLUTIONS 2003

Q1 a) $1.3274 = 1 \frac{3274}{9990} = 1 \frac{3271}{9990}$ (1)

b) 1.98 (2)

c) $\sqrt{800}$
 $= \sqrt{2 \times 400}$
 $= 20\sqrt{2}$
 $x = 20$ (2)

d) i) $-2ab + 10b$ ii) $8-8y-4+y$
 $= 4-7y$ (3)

e) $\frac{x^6 \times y^6}{x^1 \times y^4}$
 $= x^5 y^2$ (2)

f) i) $2y^2 + 10y - 3y - 15$ ii) $4x^2 + 12x + 9$
 $= 2y^2 + 7y - 15$ (2)
 $= 4x^2 + 12x + 9$ (2)

iii) $25a^2 - b^2$ iv) $6 - 9 \times 2$
 $= 6 - 18$
 $= -12$ (2)

v) $5 + 2\sqrt{10} + 2$
 $= 7 + 2\sqrt{10}$ (2)

g) $\frac{2\sqrt{5}}{5} \times \frac{\sqrt{5}}{\sqrt{5}} = \frac{2\sqrt{5}}{5}$ (2)

Q2 a) i) $2xy(2x-1)$ ii) $x(m+5) + 7(m+5)$
 $= (m+5)(x+7)$ (2)

iii) $(m-6)(m-3)$ iv) $4x^2 + 4x - 3$
 $= 4x^2 + 6x - 2x - 3$
 $= 2x(2x+3) - 1(2x+3)$
 $= (2x+3)(2x-1)$ (3)

v) $3(x^2 - 4y^2)$
 $= 3(x+2y)(x-2y)$ (2)

vi) $2^3 + (3y)^3$
 $= (2+3y)(4-6y+9y^2)$ (2)

b) i) $\frac{x^2}{x(5x-2)}$ ii) $\frac{(t+2)(t-1)}{(t+3)(t+2)}$ iii) $\frac{2x(y+1)-6(y+1)}{4(x^2-4x+3)}$
 $= \frac{x}{5x-2}$ (2)
 $= \frac{t-1}{t+3}$ (3)
 $= \frac{(2x-6)(y+1)}{4(x-3)(x-1)}$
 $= \frac{2(x-3)(y+1)}{4(x-3)(x-1)}$
 $= \frac{y+1}{2(x-1)}$ (5)

Q2 b) iv) $\frac{3}{b+2} \times \frac{b(b+2)}{3(b-1)}$
 $= \frac{b}{2b-1}$ (3)

v) $\frac{3ab^2}{5xy} \times \frac{xy(x+2y)}{6a(2b-1)}$
 $= \frac{b^2(x+2y)}{10(2b-1)}$ (3)

vi) $\frac{p-3}{6} + \frac{3p+6}{6}$ vii) $\frac{2}{(x+2)(x-2)} - \frac{3(x-2)}{(x+2)(x-2)}$
 $= \frac{4p+3}{6}$ (2)
 $= \frac{2-3x+6}{(x+2)(x-2)}$
 $= \frac{8-3x}{(x+2)(x-2)}$ (4)

c) i) $\sqrt{4 \times 3} - \sqrt{9 \times 3}$ ii) $2\sqrt{4 \times 2} - \sqrt{9 \times 2}$
 $= 2\sqrt{3} - 3\sqrt{3}$
 $= -\sqrt{3}$ (3)
 $= 4\sqrt{2} - 3\sqrt{2}$
 $= \sqrt{2}$ (3)

ii) $4\sqrt{16 \times 3} + 3\sqrt{49 \times 3} + 5\sqrt{4 \times 3}$
 $= 16\sqrt{3} + 21\sqrt{3} + 10\sqrt{3}$
 $= 47\sqrt{3}$ (5)

d) $\frac{\sqrt{3}-4}{2+3\sqrt{3}} \times \frac{2-3\sqrt{3}}{2-3\sqrt{3}}$
 $= \frac{2\sqrt{3}-9-8+12\sqrt{3}}{4-27}$
 $= \frac{14\sqrt{3}-17}{-23}$
 $= \frac{17-14\sqrt{3}}{23}$

$\therefore a = \frac{17}{23}$ $b = \frac{-14}{23}$ (5)

Q3 a) i) $6x+14=6-x+1$ ii) $3y-4 > 5y+12$
 $6x+14=7-x$
 $6x=-7-x$
 $7x=-7$
 $x=-1$ (3)

iii) $-4 \leq 2x \leq 2$ iv) $4(a-2) < 24+9a$
 $-2 \leq x \leq 1$ (2)
 $4a-8 < 24+9a$
 $4a < 32+9a$
 $-5a < 32$
 $a > \frac{32}{-5}$ (3)

v) $2x+1=3x-2$ or $2x+1=-(3x-2)$
 $2x=3x-3$ $2x+1=-3x+2$
 $-x=-3$ $2x=-3x+1$
 $x=3$ $5x=1$
 $x=\frac{1}{5}$ (4)
 (not a solution) $|\frac{1}{5}| \neq \frac{3}{5} - 2$
 $\neq -\frac{12}{5}$

vi) $2t-3 \geq 5 \leq -(2t-3) \geq 5$
 $2t \geq 8$ $2t-3 \leq -5$
 $t \geq 4$ $2t \leq -2$
 $t \leq -1$ (4)

b) i) $3a^2 - 14a + 8 = 0$ ii) $x = \frac{-12 \pm \sqrt{144 - 4 \cdot 4 \cdot 1}}{2 \cdot 4}$
 $3a^2 - 12a + 2a + 8 = 0$
 $3a(a-4) + 2(a-4) = 0$
 $(3a+2)(a-4) = 0$
 $a = \frac{2}{3}$ or $a = 4$ (4)

iii) $x = \frac{-12 \pm \sqrt{128}}{8}$
 $= \frac{-12 \pm 8\sqrt{2}}{8}$
 $= \frac{-3 \pm 2\sqrt{2}}{2}$ (4)

Q2 b) iv) $\frac{3}{b+2} \times \frac{b(b+2)}{3(b-1)}$
 $= \frac{b}{2b-1}$ (3)

v) $\frac{3ab^2}{5xy} \times \frac{xy(x+2y)}{6a(2b-1)}$
 $= \frac{b^2(x+2y)}{10(2b-1)}$ (3)

vi) $\frac{p-3}{6} + \frac{3p+6}{6}$ vii) $\frac{2}{(x+2)(x-2)} - \frac{3(x-2)}{(x+2)(x-2)}$
 $= \frac{4p+3}{6}$ (2)
 $= \frac{2-3x+6}{(x+2)(x-2)}$
 $= \frac{8-3x}{(x+2)(x-2)}$ (4)

c) i) $\sqrt{4 \times 3} - \sqrt{9 \times 3}$ ii) $2\sqrt{4 \times 2} - \sqrt{9 \times 2}$
 $= 2\sqrt{3} - 3\sqrt{3}$
 $= -\sqrt{3}$ (3)
 $= 4\sqrt{2} - 3\sqrt{2}$
 $= \sqrt{2}$ (3)

ii) $4\sqrt{16 \times 3} + 3\sqrt{49 \times 3} + 5\sqrt{4 \times 3}$
 $= 16\sqrt{3} + 21\sqrt{3} + 10\sqrt{3}$
 $= 47\sqrt{3}$ (5)

d) $\frac{\sqrt{3}-4}{2+3\sqrt{3}} \times \frac{2-3\sqrt{3}}{2-3\sqrt{3}}$
 $= \frac{2\sqrt{3}-9-8+12\sqrt{3}}{4-27}$
 $= \frac{14\sqrt{3}-17}{-23}$
 $= \frac{17-14\sqrt{3}}{23}$

$\therefore a = \frac{17}{23}$ $b = \frac{-14}{23}$ (5)

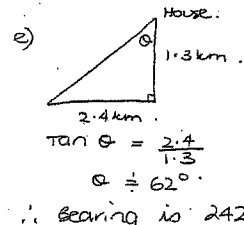
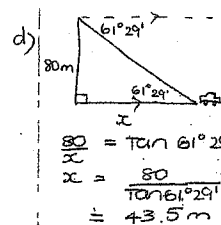
Q3 c) $4-2a-2=0$
 $2-2a=0$
 $2a=2$
 $a=1$

$\therefore x^2 - x - 2 = 0$
 $(x-2)(x+1) = 0$
 $\therefore x = 2$ and $x = -1$ (3)
 the other solution is $x = -1$

d) i) $3a+2b=5$ ii) $y=x^2+4x-1$
 $2a-b=-6$ (2)
 mult. ② by 2
 $4a-2b=-12$
 $3a+2b=5$ (+)
 $7a=-7$
 $a=-1$
 sub $a=-1$ into ①
 $3+2b=5$
 $2b=2$
 $b=1$ (4)

iii) $y=x^2+4x-1$
 $2x-y-1=0$ (2)
 sub ① into ②
 $2x-(x^2+4x-1)-1=0$
 $2x-x^2-4x+1-1=0$
 $-x^2-2x=0$
 $x^2+2x=0$
 $(x+1)(x+1)=0$
 $x=-1$
 sub $x=-1$ into ①
 $y=1-4$
 $y=-3$ (4)

Q4 a) i) 0.621 ii) -1.397 (2)
 b) i) $34^\circ 12'$ ii) $45^\circ 35'$ (2)
 c) i) $\frac{x}{10.7} = \cos 61^\circ 50'$ ii) $\sin \theta = \frac{5.8}{11.7}$
 $x = 10.7 \cos 61^\circ 50'$
 $= 5.1 \text{ cm}$ (2)
 $\theta = 29^\circ 43'$ (2)



Q5 a) i) $\frac{\sqrt{3}}{2} + \frac{1}{2}$ ii) $(\frac{1}{\sqrt{2}})^2$ iii) $\frac{2}{\sin 30^\circ}$

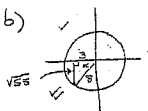
$= \frac{\sqrt{3}+1}{2}$ (3) $= \frac{1}{2}$ (2) $= \frac{2}{1/2}$ (3)

iv) $\frac{1+\sqrt{3}}{1-\sqrt{3}}$ (2) v) $-\sin 45^\circ$ vi) $\cos 30^\circ$

$= \frac{1+\sqrt{3}}{1-\sqrt{3}} \times \frac{1+\sqrt{3}}{1+\sqrt{3}}$ $= -\frac{1}{\sqrt{2}}$ (2) $= \frac{\sqrt{3}}{2}$ (2)

$= \frac{1+2\sqrt{3}+3}{1-3}$ vii) $\tan 45^\circ$ viii) $-\sin 60^\circ$

$= \frac{4+2\sqrt{3}}{-2}$ $= 1$ (2) $= -\frac{\sqrt{3}}{2}$ (2)

$= -2-\sqrt{3}$ (4) b)  $\tan x = \frac{\sqrt{55}}{3}$ (3)

i) (1st + 2nd quad) ii) (2nd + 3rd quad)

$\theta = 27^\circ 35'$ and $152^\circ 25'$ $\theta = 103^\circ 43'$ and $256^\circ 17'$

(2) (2)

iii) $\sin \theta = \frac{\sqrt{3}}{2}$ iv) $0^\circ \leq \theta \leq 720^\circ$

(1st + 2nd quad) (1st + 3rd quad)

$\theta = 60^\circ$ and 120° (2) $2\theta = 60^\circ, 240^\circ, 420^\circ, 600^\circ$

$\theta = 30^\circ, 120^\circ, 210^\circ, 300^\circ$ (3)

v) $\cos \theta = \pm \frac{1}{\sqrt{2}}$

(1st, 2nd, 3rd + 4th quad)

$\theta = 45^\circ, 135^\circ, 225^\circ, 315^\circ$ (3)

Q6 a) i) $\sin \theta = \frac{\cos \theta}{\sin \theta}$ ii) $3(1 + \tan^2 \theta)$ iii) $\cot \theta (1 - \cos^2 \theta)$

$= \frac{\cos \theta}{\sin^2 \theta}$ $= 3 \sec^2 \theta$ $= \frac{\cos \theta}{\sin \theta} \cdot \sin^2 \theta$

$= \cos \theta$ (2) $= \sin \theta \cos \theta$ (4)

b) i) $\frac{1}{\cos^2 \theta} - \sin^2 \theta$ ii) $\frac{\cos \theta (\sin \theta + \cos \theta)}{1 - \sin^2 \theta}$

$= \sec^2 \theta - \sin^2 \theta$ $= \frac{\cos \theta (\sin \theta + \cos \theta)}{\cos^2 \theta}$

$= 1 + \tan^2 \theta - \sin^2 \theta$ $= \sin \theta + \cos \theta = \tan \theta + 1$ (4)

Q6 c) i) $\frac{x}{\sin 43^\circ} = \frac{10.7}{\sin 79^\circ}$ ii) $x^2 = 8^2 + 10^2 - 2 \times 8 \times 10 \times \cos 64^\circ$

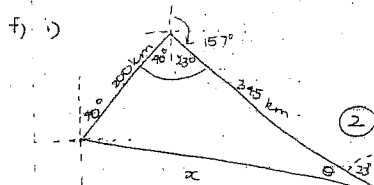
$x = \frac{10.7 \sin 43^\circ}{\sin 79^\circ}$ $x^2 = 107.2699$

$x = 7.4 \text{ cm}$ (3) $x = 10.4 \text{ m}$ (3)

d) $\cos \theta = \frac{3.6^2 + 4.2^2 - 3.8^2}{2 \times 3.6 \times 4.2}$ e) $A = \frac{1}{2} \times 1.7 \times 1.5 \times \sin 65^\circ$

$= \frac{16.16}{30.24}$ $= 1.2 \text{ m}^2$ (2)

$\theta = 57^\circ 42'$ (3)



ii) $x^2 = 200^2 + 245^2 - 2 \times 200 \times 245 \times \cos 63^\circ$

$x^2 = 96374.31104$

$x = 310 \text{ km}$ (3)

iii) $\cos \theta = \frac{310^2 + 345^2 - 200^2}{2 \times 310 \times 345}$

$= \frac{175125}{213900}$

$\theta = 35^\circ$

Bearing = $360^\circ - 23^\circ - 35^\circ = 302^\circ$ (3)

